

CHE 363, Separation Processes and Mass Transfer

Unique Number 17125

Fall 2026 – Updated 8/14/26

Lecture: MWF 10–10:50 am in CPE 2.220

Recitation: W 1–3 pm in CPE 2.216

Instructor: Dr. Ariel Z. Plantz

Office*: Main Campus CPE 3.470 (M and W)/ Pickle Research Campus EME 1.308 (T, Th, & F)

Email: ariel.plantz@utexas.edu

Office hours: M 1–2:30 pm and W 3–4 pm
and by appointment

Teaching Assistant: Eivarth Caballero

Office*: TBA

Email: Ecaballero@utexas.edu

Office hours: T & Th 1–2:30 pm
and by appointment

Grader: TBD

COURSE DESCRIPTION

Design and analysis of equilibrium and mass transfer based on separations such as absorption, stripping, distillation, extraction, and membrane-based processes. Chemical engineering majors must make a grade of at least C- in this course.

PRE-REQUISITES FOR THE COURSE

Chemical Engineering 319 (or 353) and 322 with a grade of at least C- in each.

What you Should Know from Previous Courses:

- Diagram, set up, and solve mass and energy balances (CHE 317)
- Calculate vapor-liquid and liquid-liquid equilibrium (CHE 322)
- Apply Raoult's law to calculate partial pressures (CHE 322)
- Use Fick's law and diffusion coefficients to analyze mass transfer (CHE 319/353)

LEARNING OUTCOMES

- General
 - Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
 - Define characteristics of the major separations processes used by chemical engineers

- Diffusion and Mass Transfer
 - Calculate overall mass transfer coefficients across multiple layers of resistance, based on gas or liquid
 - Predict mass transfer coefficients for packing geometries based on correlations
 - Identify dominant resistance layer limiting overall mass transfer and thus quality of the separation
- Gas/liquid separations including absorption, stripping, and distillation
 - Select and use thermodynamic and mass transfer models in Aspen
 - Use rate-based models to size these processes in Aspen
 - Use equilibrium stages to size these processes in Aspen
 - Explain trade off height/number of stages versus energy use/solvent use using AspenPlus
 - Understand limiting conditions: minimum reflux, minimum number of stages
 - Represent graphically the performance of a separating column
- Liquid/Liquid Extraction
 - Design an equilibrium cascade for LLE in Aspen.
 - Estimate minimum no of stages and minimum solvent rate
- Membrane separations
 - Understand material selection for diffusion and solubility performance
 - Explain the role of process configurations
- Adsorption
 - Compare pressure swing and temperature swing regeneration
 - Design simple system for pressure swing adsorption

EXPECTATIONS FOR SUCCESSFUL LEARNING

- The instructor will...
 - Provide lecture MWF
 - Host 3 h of weekly office hours
 - Respond in a timely manner to communications on email and canvas (canvas preferred)
 - Provide in class review prior to exams
 - Provide clear rubrics and feedback from assessments
- The TA will...
 - Teach Aspen tutorials and Aspen practice in weekly recitations
 - Host 3 h of office hours weekly
 - Make and post detailed HW/exam solutions
 - Provide clear rubrics and feedback from assessments
 - Respond in a timely manner to communications on email or Canvas

- The grader will...
 - Use detailed rubrics provided by the instructors to evaluate homeworks
 - Provide written feedback on errors
- Students will...
 - Attend lecture
 - Attend recitation
 - Attend office hours as needed
 - Complete all assigned homework
 - Work through lecture/recitation problems not completed in class
 - Read SHR or other assigned readings before lecture
 - Reach out to instructors for help when necessary
- We all will
 - Treat each other with respect, kindness, and courtesy
 - Give our best effort
 - Ask for help when needed

COMMUNICATION

The course Canvas site can be found at utexas.instructure.com. **Please message the instructors through Canvas.** You are responsible for ensuring that the primary email address you have recorded with the university is the one you will check for course communications because that is the email address that Canvas uses.

- Homework/Exam Questions
 - **Canvas discussions will be made for each assignment and exam.** Please direct course content related questions here so all students may benefit from the discussion.
 - While encouraged to use Canvas discussions, feel free to directly message the instructors as well.
- Personal Questions
 - Please message either the instructor or TA on canvas (or email) with any personal questions or concerns such as excused absences, exam conflicts, or requests to meet one-on-one. The instructors will be available for one-on-one discussions as needed to help students navigate the course successfully.

COURSE REQUIREMENTS AND GRADING

Required Textbook: Separation Process Principles with Applications Using Process Simulators, 4th edition
By Seader, Henley, and Roper (SHR)

We are participating in the Longhorn Textbook Access Program (LTA). The LTA program provides the Course Materials to students in digital format through Canvas (no physical copies will be ordered) on the first day of classes (or when you publish your Canvas course). Students have until the 12th class day to “opt-out” in Canvas and purchase an alternative elsewhere if they choose not to participate. August 31st, 2026- students will be billed on their UT “What I Owe” account for the LTA material (but they still have through the 12th class day, September 9th, to opt out). Students have up to the 20th class day to pay their LTA bill. Should students not pay their “What

I Owe” account, the charge is reversed, and students lose access. Should students not wish to participate in the program, they may opt out via Canvas and order a copy elsewhere.

Required Technology: Students are required to have a **personal laptop with Aspen Plus software**. Access to Aspen on personal computers is required for exams., but homeworks may be completed in the chemical engineering computer lab. Windows users need to download Aspen Plus and Mac users need to set up the virtual desktop to run Aspen. Mac students are required to use the virtual desktop for exams, but will only be allowed to use it in class or on exams. Mac users will need to complete homework assignments in the chemical engineering computer lab. See the following wikis for instructions on how to install Aspen Plus (Windows), the virtual desktop (Macs), and the UT VPN (Windows and Macs).

Installing Aspen V14:

<https://cloud.wikis.utexas.edu/wiki/spaces/cheit/pages/279152815/Aspen+V14+Installation>

Remote Desktop Access for Mac Users:

<https://cloud.wikis.utexas.edu/wiki/spaces/cheit/pages/66289794/Remote+Desktop+Access+for+Mac+users>

Installing VPN (Required for Off Campus Aspen Use):

<https://cloud.wikis.utexas.edu/wiki/spaces/engritgpublic/pages/81169243/Connecting+to+the+University+of+Texas+VPN>

GETTING HELP WITH TECHNOLOGY

Students needing help with technology in this course should contact the [UT Service Desk](#). For help with Aspen Plus installation first consult the Wiki and if still unable to get Aspen running message the TA. **Do not message the TA before reading through the Wiki!**

ARTIFICIAL INTELLIGENCE

The use of generative artificial intelligence tools (or Large Language Models [LLMs]) such as CoPilot or ChatGPT in this class is permitted for students who wish to use them as an aide when doing homework or studying, however, all final work submitted should be your own. AI may be useful when learning how to troubleshoot AspenPlus, however, it **will not be permitted for use on exams**. Therefore, students should only use AI to teach them how to independently run Aspen Plus.

ASSESSMENTS

The following table represents how you will demonstrate your learning and how we will assess the degree to which you have done so.

Assessments	Percent of Total Grade
1. Homework	20%
2. Midterm Exams (x3)	20% Each
3. Final	15%
4. Attendance (Completion-Based Pop Quizzes)	5%

HOMEWORKS

There are **11 homework assignments** throughout the semester that will be weighted equally in your final grade. The lowest homework will be dropped. **Late work will only be accepted within 24 h of the due date (for 60% credit).** This policy is not negotiable, as homework solutions will be posted 24 h after assignments are submitted. Any homework later than 24 h will receive a zero unless an exemption has been discussed with the instructors. **Homeworks will be due Thursday at 11:59 pm**, except exam weeks. This schedule can be changed at any time at the instructors discretion.

EXAMS

This course will have three midterms and a final exam. **Midterms will be three hours 6-9 pm on the Monday after the review week.** This date is subject to change. Students are expected to immediately alert the instructor of any conflicts, including course conflicts. Failure to alert the instructors of a known conflict at least two weeks prior to an exam may result in a penalty. Other excused absences such as illness or family emergency should be discussed with the instructor as soon as possible so alternative arrangements can be made. The final exam time is set by the university and cannot be changed. If you cannot attend the final, you cannot take this course.

Final Exam Schedule: <https://onestop.utexas.edu/registration-and-degree-planning/final-exam-information/>

Each exam will contain a closed-note book, conceptual based portion and an open-note Aspen-based portion (except the final). Students are required to have a personal laptop which can run Aspen Plus for each exam. The use of AI or any messaging platforms (email, text, FB messenger, whatsapp, etc) will not be permitted. Any use of prohibited services may result in a zero on the exam and potential referral to the student conduct office. Exams are not cumulative, but the material inherently builds on concepts from previous sections. The closed-note, conceptual portion of the final will be cumulative.

ATTENDANCE

The instructor will periodically provide in class pop quizzes in lecture and/or recitation in order to gauge understanding and track attendance. Quizzes **will be graded on completion only**. Please alert the instructors of any excused absences prior to class on Canvas (or email). The frequency of these quizzes is up to the instructors discretion. **Students are not expected to attend in person if they are sick**, have a job interview, attend a career fair, or have a family emergency. Other excused absences include observance of religious holy days (see below). Please communicate these excused absences to the instructor via Canvas (or email) prior to the absence.

RELIGIOUS HOLY DAYS

By UT Austin policy, you must notify the instructors of your pending absence for a religious holy day as far in advance as possible of the date of observance. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence. For questions regarding religious accommodations, please contact the Office of the Dean of Students.

EXTRA CREDIT

Extra credit may be available at the instructors discretion.

CLASS RECORDINGS AND HANDOUTS

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings. HOP 2-9970 prohibits students from recording class instruction (audio or video) unless a student obtains the instructor's permission or Disability & Access has approved audio recording as an accommodation.

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class without my explicit, my written permission. Unauthorized sharing of materials may facilitate cheating. The University is aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

GRADING POLICY

+/- grades will be used for the final class grade.

Grade	Cutoff
A	94%
A-	90%
B+	87%
B	84%
B-	80%
C+	77%
C	74%
C-	70%
D+	67%
D	64%
D-	60%
F	<60%

ACADEMIC INTEGRITY EXPECTATIONS

Students are encouraged to work together on assignments, but all submitted work should be your own. All allegations of academic misconduct will be referred by faculty to the Office of the Dean of Students. Students who violate University rules on academic misconduct are subject to the student conduct process. A student found responsible for academic misconduct may be assigned both a status sanction and a grade impact for the course. The grade impact could range from a zero on the assignment in question up to a failing grade in the course. A status sanction can include a written warning, probation, deferred suspension, suspension, or expulsion from the University. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website at: <http://deanofstudents.utexas.edu/conduct>.

TITLE IX DISCLOSURE

Texas Education Code, Section 51.252 requires all employees of Texas universities, including faculty, to report to the Title IX Office any information regarding incidents of sexual harassment, sexual assault, dating violence, or stalking that is disclosed to them. Texas law requires that all employees who witness or receive information about incidents of this nature (including, but not limited to, written forms, applications, one-on-one conversations, class assignments, class discussions, or third-party reports) must report it to the Title IX Coordinator. Before talking with any faculty or staff member about an incident of this nature, please be aware that they are required to report that information. Graduate teaching and research assistants are also designated as mandatory reporters and are required to report behaviors that are considered sexual misconduct, including the types of misconduct described above. Pregnant and parenting students are also eligible to receive support and resources from the Student Parent Liaison in the Title IX Office. Support and resources may include supportive measures, medically necessary periods of leave, and leaves of absence. Case managers from the Title IX Office connect with students to provide one-on-one, individualized support. If you would like to speak with a case manager for support, please email: supportandresources@austin.utexas.edu. For more information about reporting options and resources, please visit the Title IX website, contact the Title IX Office via email at: titleix@austin.utexas.edu, or call 512-471-0419.

For a list of important university policies and helpful resources that you may need as you engage with and navigate your courses and the university, see the [University Policies and Resources Students Canvas](#) page. The page includes the language of the University Honor Code, Title IX legal requirements for Texas employees, and information about how to receive support through the office of Disability & Access.

COURSE OUTLINE

All instructions, assignments, readings, rubrics and essential information will be on the Canvas website at utexas.instructure.com. Check Canvas regularly. **Changes to the schedule may be made at the instructors discretion if circumstances require, including the lecture schedule, homework due dates, and exam dates.** The instructor will announce any such changes in class and will also communicate them via a Canvas announcement. It is your responsibility to note these changes when announced, and the instructor will do their best to ensure that you are notified of changes with as much advance notice as possible.

*Aspen Plus Required

Wk	Date	Readings	Class Topic	Assigned	Due
1	8/24	SHR 1, 7 ChE Seps, SHR 2 & 4.3	Introduction and Flash Calculations	*HW 1	
2	8/31	SHR 3.1–5, 3.7, 6.1.2–3, Song	Mass Transfer	HW 2	*HW1
3	9/7	SHR 6.3, 6.7, SHR 6.8.1–2	Rate-Based Absorption	*HW 3	HW 2
4	9/14		Exam 1 Review		*HW3
5	9/21	SHR 6.1.1, 6.6	Stripping and Trays	*HW 4	*Exam 1
6	9/28	SHR 6.3, 6.5	Equilibrium Stages and Reboiled Strippers	*HW 5	*HW 4
7	10/5	SHR 7.0–2, 9.1.3, 9.1.6	Binary Distillation	*HW 6	*HW 5
8	10/12		Exam 2 Review		*HW 6
9	10/19		Rating Models	*HW 7	*Exam 2
10	10/26	SHR 9.1 Pre-Recorded Lectures MWF	Multicomponent Distillation	*HW 8	*HW 7
11	11/2	SHR 8 & 10.8	Liquid Liquid Extraction	*HW 9	*HW 8
12	11/9	SHR 15	Exam 3 Review/Adsorption		*HW 9
13	11/16	SHR 14 Pre Recorded Lectures WF	Adsorption/Membranes	HW 10	*Exam 3
14	11/30	TBD	Membranes	HW 11	HW 10
15	10/7	TBD	Special Topics/Final Review		HW 11, *Final